

20th Jan 2026 | DA6360

→ Formulate & Categorize

- Let's say we are given a set of points $(\alpha_i)_{i=1}^n$ in the real line $(\mathbb{R})$ find a pt. $x \in \mathbb{R}$ s.t. sum of the squared distances from x to each α_i is minimised.

$$f(x) = \sum_{\alpha=1}^n (x - \alpha_i)^2 = \sum_{\alpha=1}^n f_i(x)$$

$$f_i(x) = (x - \alpha_i)^2$$

$$f(x) = \sum_{\alpha=1}^n (x - \alpha_i)^2$$

$$= \sum_{\alpha=1}^n x^2 - 2\alpha_i x + \alpha_i^2$$

$$= nx^2 - 2x \sum_{\alpha=1}^n \alpha_i + \sum_{\alpha=1}^n \alpha_i^2$$

$$f(x) = ax^2 + bx + c$$

$$= a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right)$$

$$= a \left(\left(x + \frac{b}{a} \right)^2 + \left(\frac{c}{a} - \frac{b^2}{4a^2} \right) \right)$$

} indi.
losses
are
quadratic

$$x_* = \frac{-b}{2a}$$

$$a = n, \quad b = -2 \sum_{\alpha=1}^n \alpha_i, \quad x_* = \frac{\sum_{\alpha=1}^n \alpha_i}{n}$$

- the above was a kind of quadratic objective & unconstrained problem.

- Same problem like above but $(\alpha_i)_{i=1}^n \in \mathbb{R}^2$

$$f(x) = \sum_{i=1}^n [(x_1 - \alpha_{i1})^2 + (x_2 - \alpha_{i2})^2]$$

$$= n(x_1^2 + x_2^2) - 2 \sum_{i=1}^n x_1 \alpha_{i1} + x_2 \alpha_{i2}$$

$$\begin{aligned} & \underbrace{x^T x}_{x^T x} + \underbrace{\sum_{i=1}^n \alpha_{i1}^2 + \alpha_{i2}^2}_{\alpha_i^T \alpha_i} - \underbrace{2 \sum_{i=1}^n x_1 \alpha_{i1} + x_2 \alpha_{i2}}_{x^T \alpha_i} \end{aligned}$$

guess the
minimiser
& minimum

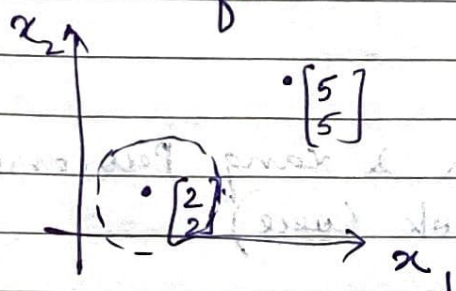
$$= n x^T x - 2 x^T \sum_{i=1}^n \alpha_i + \sum_{i=1}^n \alpha_i^T \alpha_i$$

$$= n \left[\left(x - \sum_{i=1}^n \frac{\alpha_i}{n} \right)^T \left(x - \sum_{i=1}^n \frac{\alpha_i}{n} \right) \right]$$

$$+ \sum_{i=1}^n \frac{\alpha_i^T \alpha_i}{n} - \left(\sum_{i=1}^n \frac{\alpha_i}{n} \right)^T \left(\sum_{i=1}^n \frac{\alpha_i}{n} \right)$$

the above one was also an eq. of quadratic objective with unconstrained problem.

- come with a rope of len 1m. Same food at $\begin{bmatrix} 5 \\ 5 \end{bmatrix}$. Find $x \in \mathbb{R}^2$ s.t. it is closest to food.



Quadratic objective

Convex Set
Constrained Problem

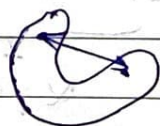
$$\min (x_1 - 5)^2 + (x_2 - 5)^2$$

$$\text{s.t. } \|x - \begin{bmatrix} 2 \\ 2 \end{bmatrix}\|_2 \leq 1$$

One more possibility $x_1 \leq 2.5$

- Convex set $\Rightarrow$ within a set, two pts should be able to see each other.

not convex



$\rightarrow$ LR Linear Reg.

given a set of pts. $(a_i, b_i)_{i=1}^n$ $a, b \in \mathbb{R}$

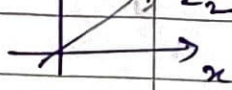
we want
a line

$$b = x_1 a + x_2$$

→ try to visualize this

— / — / —

$f(x)$ this is line, we want a linear model.

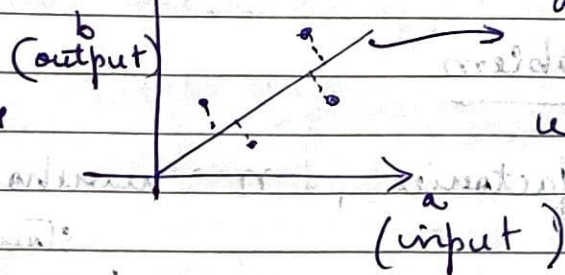


$$f(x) = \sum_{i=1}^n (b_i - \underbrace{[x_1 a_i + x_2]}_{\text{predicted value}})^2$$

true value

$$\min_{x_1, x_2} f(x)$$

objective function



a model gives this line for a particular value of x_1 & x_2 .

- Package the same in vector form

$$\text{error} = \begin{bmatrix} \text{error}(1) \\ \vdots \\ \text{error}(n) \end{bmatrix} \in \mathbb{R}^n = \begin{bmatrix} b_1 - (x_1 a_1 + x_2) \\ \vdots \\ b_n - (x_1 a_n + x_2) \end{bmatrix}$$

$$f(x) = \|\text{error}\|_2^2$$

$$A = \begin{bmatrix} a_1 & \dots & a_n \\ 1 & \dots & 1 \end{bmatrix}_{2 \times n}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

now my error, becomes $b - A^T x$

can be rewritten as -

$$f(x) = [b - (A^T x)]^T [b - (A^T x)]$$

$$\min_x f(x) = \min_x \|b - A^T x\|_2^2$$

Quadratic Eqⁿ
Objective

- say I want to be sure to be only positive

$$\min \|b - A^T x\|_2^2$$

$$\text{s.t. } A^T x \geq b \quad (\text{it has } n \text{ constraints})$$

→ Transportation Problem

$m \rightarrow$ plants / factories, $n \rightarrow$ warehouses / storage facility
(produced) (stored)

$x_{ij} \rightarrow$ goods transported from plant i to warehouse j .

$c_{ij} \Rightarrow$ associated transported cost.

$s_i, i=1, \dots, m$ (supply)

$d_j = j=1, \dots, n$ (demand)

The goal is to satisfy the demand at the min cost.

$$\text{Obj.} \rightarrow \sum_{j=1}^n \sum_{i=1}^m c_{ij} x_{ij} \quad [\text{trace}(C^T x)]$$

$$\text{s.t. } \sum_{i=1}^m x_{ij} = d_j$$

inequality constraints $\left\{ \begin{array}{l} \sum_{j=1}^n x_{ij} \leq s_i \\ x \geq 0 \end{array} \right.$

⇒ Order Statistics

(given) $\Rightarrow n$ nos. $\rightarrow c_1, \dots, c_n \rightarrow$ given array

then order $\Rightarrow c_{(1)} \leq c_{(2)} \leq \dots \leq c_{(n)}$

(remember this) $c_9 \leq c_5 \dots$ could be

Find the sum of first k nos. (after sort).
 $(c_{(1)} + \dots + c_{(k)})$

not chosen $\rightarrow$ chosen
 $x_i \in \{0, 1\}$

kind of
a discrete
optimization
problem

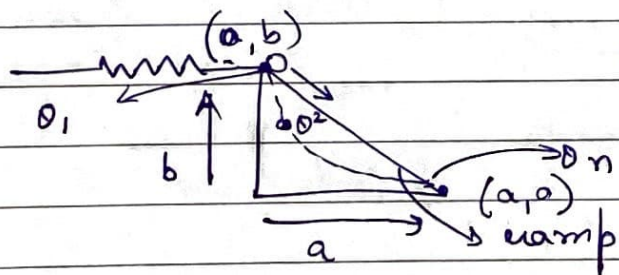
discrete
var.

min $\sum_i x_i c_i \rightarrow$ linear objective
 s.t. $\sum_{i=1}^n x_i = k \rightarrow$ linear constraint

Eg. $c_i: 2, 5, 3, 8, 1$; $k = 3$
 $x_i: 1, 0, 1, 0, 1$

} Tabular
Method of
Programming.

⇒ Brachistochrone Problem



Ramp: $y(x)$
 velocity: $v(x)$

$$mgb = mgy(x) + \frac{1}{2}mv(x)^2$$

$$g(b - y(x)) = \frac{1}{2}v(x)^2$$

$$v(x) = \sqrt{2g(b - y(x))}$$

- Time spent in i^{th} piece

$$t^i = \frac{\|o^i - o^{i-1}\|}{v^i}$$

$$v^i = \sqrt{2g \left(b - \frac{o^i + o^{i-1}}{2} \right)}$$

$$\min_{i=1}^N t^i$$

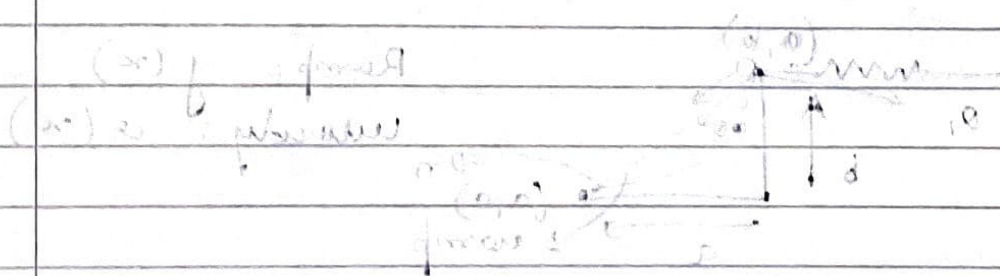
non-linear,
non-convex

it has n variables

$$\text{s.t. } \begin{cases} o_1^i \geq o_1^{i-1} \\ o_2^i \leq o_2^{i-1} \end{cases}$$

linear constraints

Practical constraints



$$f(x) = \begin{cases} m_1(x-a) + d_1 & a \leq x < x_1 \\ m_2(x-x_1) + d_2 & x_1 \leq x < x_2 \\ \vdots \\ m_n(x-x_{n-1}) + d_n & x_{n-1} \leq x \leq b \end{cases}$$

$$g(x) = \begin{cases} d_1 & a \leq x < x_1 \\ d_2 & x_1 \leq x < x_2 \\ \vdots \\ d_n & x_{n-1} \leq x \leq b \end{cases}$$

$$f(x) = g(x) \quad \forall x \in [a, b]$$